Experiment - 04

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Semester: 5
Date of Performance: 27/7/26
Subject Name: ADBMS
Subject Code: 24CSP-310

(A)

(i) You are given a table tbl_happiness that contains the ranking of happiness for different countries. Your task is to retrieve all the records from the table while ensuring that "India" and "Sri Lanka" appear at the top, followed by the rest of the countries in their original ranking order. Output:

Rankings	Country
126	India
128	Sri Lanka
1	Finland
2	Denmark
3	Iceland
4	Israel
5	Netherlands
6	Sweden
7	Norway

Solution:

```
SELECT RANKINGS, COUNTRY,
CASE
WHEN COUNTRY='India' THEN 1
WHEN COUNTRY='Sri Lanka' THEN 2
ELSE 3
END AS DERIVED_RANK
FROM TBL_HAPPINESS
```

ORDER BY DERIVED_RANK, RANKINGS

(B)

(ii) Make all the rankings in descending order except India and Sri Lanka. Output:

Rankings	Country
126	India
128	Sri Lanka
7	Norway
6	Sweden
5	Netherlands
4	Israel
3	Iceland
2	Denmark
1	Finland

Solution:

```
SELECT RANKINGS, COUNTRY,
CASE
WHEN COUNTRY='India' THEN 1
WHEN COUNTRY='Sri Lanka' THEN 2
ELSE 3
END AS DERIVED_RANK
FROM TBL_HAPPINESS
ORDER BY DERIVED_RANK, RANKINGS DESC;
```

(C)

A company maintains employee information in a single table where each employee may report to another employee acting as their manager. Prepare a report that displays the details of every employee who supervises at least one other employee.

The report should include the manager's employee ID, name, total number of direct reports, and the average age of those reporting employees. The average age should be rounded to the nearest whole number and the final result should be sorted by employee ID.

employee_id	name	reports_to	age
9	Hercy	NULL	43
6	Alice	9	41
4	Bob	9	36
2	Winston	NULL	37

Output

employee_id	name	reports_count	average_age
9	Hercy	2	39

Solution:

```
SELECT
E1.EMPLOYEE_ID, E1.NAME, COUNT(E2.REPORTS_TO) AS
REPORTS_COUNT, ROUND(AVG(E2.AGE)::NUMERIC,
0) AS AVERAGE_AGE
FROM EMPLOYEES AS E1
JOIN EMPLOYEES AS
E2
ON E1.EMPLOYEE_ID = E2.REPORTS_TO
GROUP BY E1.EMPLOYEE_ID, E1.NAME
ORDER BY E1.EMPLOYEE_ID;
```